

Mohamed Abdelwahab, PhD

Senior Embodied AI Engineer,
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Profile Summary

Results-driven British Egyptian engineer and researcher with a proven track record of advancing R&D initiatives and driving enterprise development. Recognised for consistently achieving deliverables in the dynamic fields of computer vision and robotic solutions across diverse applications. I specialise in developing and implementing innovative ideas and excel in solving real-world challenges. I am passionate about collaborating with diverse groups and am energised by the opportunity to push boundaries, consistently delivering impactful outcomes.

Key Achievements

- **Enhanced Robot Navigation & Perception:** Integrating bio-inspired camera system for enhanced visual SLAM, optimising human detection using active perception for elder monitoring robots
- **Optimised Computer Vision Solutions:** Crafting innovative solutions using computer vision and deep learning across diverse applications
- **Industrial Inspection Robotics:** Designing autonomous inspection robots tailored for industrial plant applications
- **Successful Team Leadership:** Successfully supervised multidisciplinary teams, resulting in outstanding project outcomes

Work Experience

May 2025 – **Senior Embodied AI Engineer, MBZUAI, Abu Dhabi, UAE**

- Present Lead technical robotics lab setup and infrastructure development for embodied AI research
- Support and collaborate with researchers across diverse robotics application domains
- Plan and coordinate research projects spanning multiple embodied AI themes

May 2024 – **Marine Robotics Autonomy Research Lead, Khalifa University, Abu Dhabi, UAE**

- May 2025 Develop navigation systems for underwater mobile robots
- Develop deep learning models for coral reef inspection
- Collaborate with the agriculture robotics team to develop robot navigation systems

- June 2023 – **Senior Computer Vision Engineer**, *Mimetrik Solutions*, Leeds, UK
- May 2024
- Developed precision-focused computer vision and deep learning solutions for digital dentistry
 - Spearheaded a web-based deep learning application, optimizing the digital dentistry workflow
 - Designed and implemented automated solutions for efficient dataset operations
- Feb 2022 – **Computer Vision Engineer**, *GroundWOW*, Manchester, UK
- June 2023
- Engineered and validated visual navigation systems, elevating the performance of ground printing robots
 - Conducted comprehensive market research for sensor selection, optimising the robots' sensing capabilities
 - Led 3D-Lidar robot localisation project in challenging environments, pushing robotic capabilities
- July 2020 – **Research Fellow in Offsite Logistics and AI**, *UWE Bristol*, Bristol, UK
- Feb 2022
- Engineered an innovative AI-cloud-IoT platform for offsite logistics, optimising efficiency and connectivity
 - Integrated Building Information Modelling (BIM) and SLAM algorithms, ensuring precise indoor robot navigation
 - Optimized asset delivery routes considering carbon footprint, cost, and disruption to road users
 - Developed a mobile application for streamlined asset data scanning
- Jan 2018 – **Research Fellow in Industrial Robotic Inspection**, *University of Surrey*, Guildford, UK
- July 2020
- Designed and developed inspection robots tailored for unmanned industrial plants, emphasising remote operation
 - Implemented deep learning for equipment detection, valve state detection, and meter reading
 - Contributed to the design of the industrial Sixth Sense (6S) framework for proactive monitoring and control
 - Supervised final year projects on RGB-D SLAM
- Nov 2016 – **Research Fellow in Elderly-Monitor Robotic System**, *University of Lincoln*, Lincoln, UK
- Nov 2017
- Developed active-perception-based human detection for health monitoring robotic systems
 - Delivered invited talks on computer vision for M.Sc. students

Education

- Feb 2012 – **PhD in Robotics (Joint Program)**, *EJUST, Egypt & Osaka University, Japan*
- Feb 2015 **Thesis:** Development of Camera Oscillation System to Improve Visual Mono SLAM Performance
- Feb 2010 – **MSc in Robotics**, *EJUST*, Egypt
- Feb 2012 **Thesis:** Study of the Effect of Feature Geometric Composition on Visual Mono SLAM Systems
- Sep 2003 – **BSc Computer Science (First Class Honours)**, *Sohag University*, Egypt
- Jul 2007 **Project:** Face Recognition Using C++ and OpenCV

Technical Skills

Core Competencies

- Multi-disciplinary Problem Solving
- Market Research & Analysis
- Team Leadership & Collaboration
- Creative Solution Development

Robotics & AI

- C++, Python, MATLAB
- SLAM & Navigation
- ROS1, ROS2, Gazebo
- Sensor Integration & Hardware

Computer Vision & Machine Learning

- PyTorch, TensorFlow
- OpenCV, Scikit-learn
- CUDA Programming
- Deep Learning Architectures

Software Development & Data Management

- Flask, Bootstrap
- MySQL, SQL Server
- Cloud Platforms, GitHub
- FME Desktop & Server

Selected Publications

- **M. Heshmat**, L. Saad Saoud, M. Abujabal, A. Sultan, M. Elmezain, L. Seneviratne, and I. Hussain, "Deep learning for underwater SLAM: A survey of recent advances and challenges," *Sensors*, 2025
- M. Elmezain, L. Saad Saoud, A. Sultan, **M. Heshmat**, L. Seneviratne, I. Hussain, "Advancing Underwater Vision: A Survey of Deep Learning Models for Underwater Object Recognition and Tracking," *IEEE Access*, 2025
- L. Saad Saoud, M. Elmezain, A. Sultan, **M. Heshmat**, L. Seneviratne, I. Hussain, "Seeing through the haze: A comprehensive review of underwater image enhancement techniques," *IEEE Access*, 2024
- L. Saad Saoud, A. Sultan, M. Elmezain, **M. Heshmat**, L. Seneviratne, I. Hussain, "Beyond observation: Deep learning for animal behavior and ecological conservation," *Ecological Informatics*, 2024
- B. Dorneanu, S. Zhang, H. Ruan, **M. Heshmat**, R. Chen, V. S. Vassiliadis, H. Arellano-Garcia, "Big data and machine learning: A roadmap towards smart plants," *Frontiers of Engineering Management*, 2022
- **M. Heshmat**, Y. Gao, "Equipment Detection based Inspection Robot for Industrial Plants," *Towards Autonomous Robotic Systems Conference (TAROS)*, 2021
- A. Mohamed, H. Ruan, **M. Heshmat**, B. Dorneanu, P. Xiao, H. Arellano-Garcia, Y. Gao, R. Tafazolli, "An Inter-disciplinary Modelling Approach in Industrial 5G/6G and Machine Learning Era," *IEEE ICC 2020 Workshop*, 2020
- **M. Heshmat**, M. Fernandez-Carmona, Z. Yan, N. Bellotto, "Active Human Detection with a Mobile Robot," *UK-RAS Conference on Robotics and Autonomous Systems*, 2017
- **M. Heshmat**, M. Abdellatif, K. Nakamura, A. A. Abouelsoud, N. Babaguchi, "Camera Oscillation Pattern for VSLAM: Translational Versus Rotational," *IEEE International Conference on 3D Imaging (IC3D)*, 2014
- **M. Heshmat**, M. Abdellatif, H. Abbas, "Improving Visual SLAM Accuracy Through Deliberate Camera Oscillations," *IEEE International Symposium on Robotics and Sensors Environments (ROSE)*, 2013